

Recurrence Relations

Example:

the rule for finding a term is defined using preceding terms

$$a_n = a_{n-1} - a_{n-2}$$

$$a_{n-1} = a_{n-2} - a_{n-3}$$

$$a_{n-2} = a_{n-3} - a_{n-4}$$

$$a_{n-3} = a_{n-4} - a_{n-5}$$

$$\begin{array}{l}
 a_4 = a_3 - a_2 \\
 = -3 - 2 \\
 = -5
 \end{array}
 \left|
 \begin{array}{l}
 a_5 = a_4 - a_3 \\
 = -5 - (-3) \\
 = -2
 \end{array}
 \right|
 \begin{array}{l}
 a_0 = 3 \\
 a_1 = 5
 \end{array}
 \left|
 \begin{array}{l}
 a_2 = a_1 - a_0 \\
 = 2
 \end{array}
 \right|
 \begin{array}{l}
 a_3 = a_2 - a_1 \\
 = -3
 \end{array}$$

$$a_5 = a_4 - a_3 = (a_3 - a_2) - (a_2 - a_1)$$

$$\begin{aligned}
 &= (\cancel{a_3} - a_2) - (\cancel{a_2} - a_1) \\
 &= -a_2 + a_1 \\
 &= -5 + 3 \\
 &= -2
 \end{aligned}$$

Modelling using recurrence relations

Q. Compound interest

Initial deposit: Rs. 10,000

interest: 11% per year compounded annually

$$a_0 = 10,000$$

$$a_1 = (0.11)a_0 + a_0 = (1.11)a_0$$

$$a_n = (1.11)^n a_0$$

$$a_0 = 10,000$$

Q. Rabbit puzzle

Initially 1 young pair of rabbits.

After 2 months, a young pair produces another pair of rabbits every subsequent month. How many pairs of rabbits are there after n months?

Soln.

$$a_0 = 1$$

$$a_1 = 1$$

$$a_2 = 2 = a_1 + a_0$$

a_i denotes no. of pairs of rabbits at the end of i months

$$a_n = a_{n-1} + a_{n-2}$$

newly created pairs.

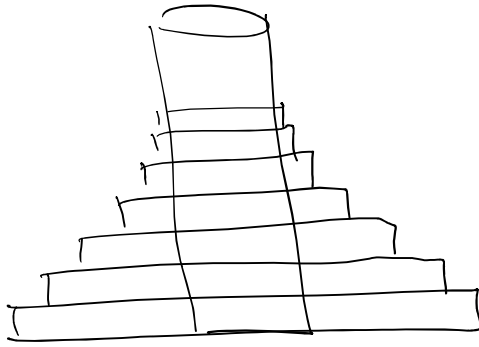
$$a_0 = 1$$

$$a_1 = 1$$

Q. Tower of Hanoi

Goal: Move everything from P₀ to P₂

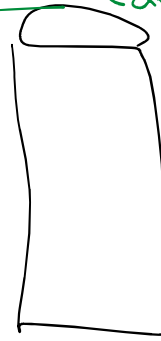
4. Tower of Hanoi



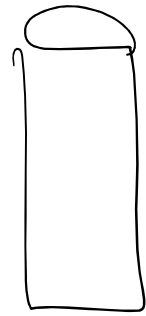
Peg 1

Goal: Move everything from Peg 1 to Peg 2.

Rule: Cannot keep a larger object on top of a smaller object.



Peg 2



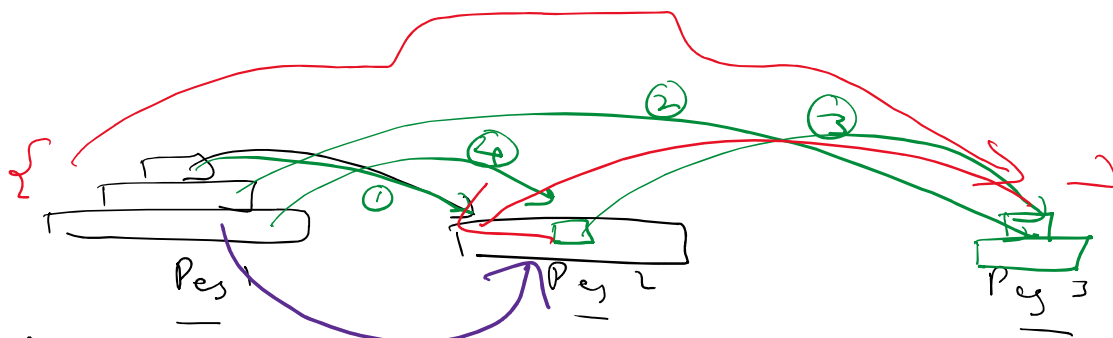
Peg 3

Let H_n denote the no. of moves needed to solve the Tower of Hanoi problem. Find a recurrence relation for H_n .

$$H_1 = 1$$

$$H_2 = 3$$

$$H_3 =$$



After 4 steps

Peg 1

Peg 2

Peg 3

$$H_3 = H_2 + 4$$

$$H_3 = H_2 + 1 + H_2$$

$$H_n = 2H_{n-1} + 1$$

$$H_1 = 1$$

$$2 \cdot 2 \cdot H_{n-2}$$

⋮

$$2^{n-1} H_1$$